The subcaptionlite package

Subfigures and subtables without the caption package

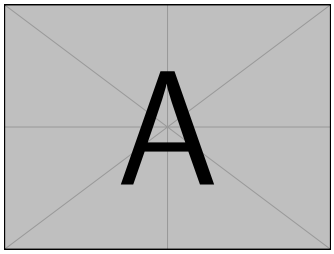
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https://github.com/munepi/subcaptionlite

v1.0 (2026/10/26)

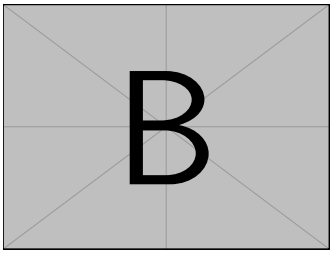
1 Introduction

The subcaption package needs the caption package, which redefines `\@makecaption` of the class. Thus it cannot be used with a class that has its own design of captions: loading it changes the captions of the whole document.

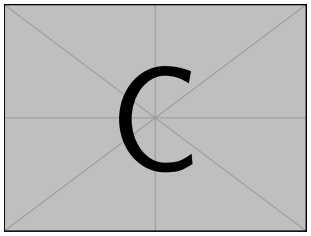
The subcaptionlite package provides the `subfigure` and `subtable` environments, `\subcaption`, `\subcaptionbox` and `\subref` with the same syntax as subcaption, without touching `\@makecaption` of the class. Figure 1 and Table 1 are made with it.



(a) The subfigure environment



(b) `\subcaptionbox` with a width



(c) `\subcaptionbox` without a width

Figure 1: Three subfigures

Table 1: Two subtables

(a) The first

Item	Value
A	1
B	2

(b) The second

Item	Value
C	3
D	4

Supported formats/engines pdfLaTeX, XeLaTeX, LuaLaTeX (with or without LuaTeX-ja), pLaTeX and upLaTeX (with dvipdfmx). The package requires LaTeX2e 2020-10-01 or later and loads etoolbox.

2 Usage

```
\usepackage{subcaptionlite}
```

```
\begin{figure}
  \centering
  \begin{subfigure}[b]{.3\linewidth}
    \centering\includegraphics[width=.9\linewidth]{example-image-a}
    \caption{The subfigure environment}\label{fig:example-a}
  \end{subfigure}\hfill
  \subcaptionbox{\cs{subcaptionbox} with a width\label{fig:example-b}}%
    [.3\linewidth]{\includegraphics[width=.9\linewidth]{example-image-b}}\hfill
  \subcaptionbox{\cs{subcaptionbox} without a width\label{fig:example-c}}{%
    \includegraphics[width=.25\linewidth]{example-image-c}}
  \caption{Three subfigures}\label{fig:example}
\end{figure}

\begin{table}
  \centering
  \caption{Two subtables}\label{tab:example}
  \begin{subtable}[t]{.3\linewidth}
    \centering
    \caption{The first}\label{tab:example-a}
    \begin{tabular}{lr} ... \end{tabular}
  \end{subtable}\hspace{2em}
  \begin{subtable}[t]{.3\linewidth}
    \centering
    \caption{The second}\label{tab:example-b}
    \begin{tabular}{lr} ... \end{tabular}
  \end{subtable}
\end{table}
```

\begin{subfigure}[⟨*pos*⟩]{⟨*width*⟩}, \begin{subtable}[⟨*pos*⟩]{⟨*width*⟩} A `minipage` of ⟨*width*⟩ (⟨*pos*⟩: `c` by default), inside which `\caption` and `\caption*` mean `\subcaption` and `\subcaption*`. The subcaption is put where `\caption` is given: above the contents in Table 1.

\subcaption{⟨*text*⟩}, \subcaption*{⟨*text*⟩} A numbered or unnumbered subcaption, e.g. in a `minipage`. It is centered and set in `\small`. It is numbered with the counter of the float it is in: `subfigure` in a `figure`, `subtable` in a `table`.

\subcaptionbox{⟨*caption*⟩}[⟨*width*⟩]{⟨*contents*⟩} Puts ⟨*contents*⟩ centered in a `minipage` of ⟨*width*⟩, followed by `\subcaption{`⟨*caption*⟩`}`. Without ⟨*width*⟩, the natural width of ⟨*contents*⟩ is used; with ⟨*width*⟩, `\linewidth` in ⟨*contents*⟩ is ⟨*width*⟩. If ⟨*contents*⟩ is wider than ⟨*width*⟩, the `minipage` keeps ⟨*width*⟩. Boxes are aligned at the bottom, and the subcaption is always below the contents.

\subref{⟨*label*⟩}, \subref*{⟨*label*⟩} Only the label of the subfigure or subtable, e.g. "(a)" (`\subref*`: without a link with hyperref). After a numbered subcaption, `\label{`⟨*label*⟩`}` also writes the label `sub@`⟨*label*⟩, as the subcaption package does.

\captionsetup[⟨*type*⟩]{⟨*options*⟩} Accepted and ignored, so that the settings for the caption package can be left in the document.

Numbering and references The counter `subfigure` is reset by `figure`, and `subtable` by `table`. `\thesubfigure` is `(\alph{subfigure})` and `\p@subfigure` is `\thefigure` (likewise for `subtable`), so that `\ref` gives "1(a)" and "1(b)", and `\Cref` of cleveref gives "Figure 1(b)" and "Table 1(a)". With hyperref, each subfigure and subtable has its own anchor (`\theHsubfigure`, `\theHsubtable`).

The `\caption` of the float may come after the subcaptions (as in Fig. 1) or before them (as in Table 1). When a subcaption comes first, it steps the counter of the float in advance and the `\caption` that follows

does not step it again, as the subcaption package does; without this, the subfigures would get the number of the previous figure. A float with subcaptions and no `\caption` is numbered as well.

3 Customization

The following macros can be redefined in the class or in a local style (`\makeatletter` is needed in the preamble of a document).

Macro	Default
`\subcaption@font`	`\small`
`\subcaption@sep`	the space between the label and the text: `\hspace{.5zw}` with pTEX, `.5\zw` with LuaTEX-ja (or a class that defines `\zw`), `.5em` otherwise
`\subcaption@skip`	the space between the subfigure and its subcaption: `\vskip.25\baselineskip`

The engine is checked when the subcaption is typeset, so LuaTEX-ja may be loaded after subcaptionlite. The labels themselves are `\thesubfigure` and `\thesubtable`. For example,

```
\makeatletter
\renewcommand*\subcaption@font{\footnotesize\sffamily}
\renewcommand*\subcaption@sep{\enspace}
\makeatother
\renewcommand\thesubfigure{\alph{subfigure})}
```

4 Errors and limitations

`\subcaption outside float` `\subcaption` (or `\subcaptionbox`) is used outside a float.

`Float type '`⟨*type*⟩**`' has no subcaptions`** `\subcaption` (or `\subcaptionbox`) is used in a float other than `figure` and `table`, e.g. one defined by newfloat.

After either error, the subcaption is typeset without a number.

- Subcaptions are for figures and tables only.
- subcaptionlite cannot be used together with the caption or subcaption package.

5 Tests

`make test` compiles the test files (but `test-errors.tex`) twice with pdfLaTeX, XeLaTeX, LuaLaTeX, pLaTeX and upLaTeX (+ dvipdfmx) and checks the log. The helpers in `test-common.tex` check the label that `\label` would write, the references, the counters, the widths of the boxes, the anchors of hyperref, and that `\@makecaption` of the class is unchanged; the log must have `SCL-RESULT OK` and no error or warning. For `test-errors.tex`, the log must have the expected error message.

Test	What it covers
`test-basic.tex`	`subfigure`, `subtable`, `\subcaption`, `\subcaptionbox`, the caption of the float below/above the subcaptions or none, `figure*`, `\ref`, `\subref`, the label separator
`test-hyperref.tex`	`test-basic.tex` with hyperref (loaded after and before the package) and cleveref
`test-japanese.tex`	Japanese classes (ltjsarticle, jsarticle, ujarticle), LuaTEX-ja loaded after the package
`test-errors.tex`	the error messages

6 History

v1.0 (2026/10/26) First public release. Support every engine; add the `subtable` environment, subcaptions in tables and `\subref`; fix the figure number when the caption of the figure comes before the subcaptions, and after a figure whose subfigures follow its caption.

7 License

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